



Trigger Definition in SQL

A **Trigger** in SQL is a special stored program that automatically executes when a specific event occurs on a table.

Triggers are activated by database events such as:

- INSERT
- UPDATE
- DELETE

Triggers help automate tasks, enforce business rules, maintain audit logs, improve security, and preserve data integrity.

Unlike stored procedures, triggers run automatically without being manually called.

Short Notes on the Scripts

SECTION 1: Introduction to Triggers

```
USE techsphere_solutions;
```

What it does

- Selects the techsphere_solutions database.
- Ensures all subsequent operations run inside the correct database.

Purpose

- Sets the working environment for the trigger demonstrations.

SECTION 2: Create Audit Tables

Script 1: Create Audit Log Table

```
CREATE TABLE IF NOT EXISTS employee_audit_log
```

What it does

Creates a table used to store employee activity logs.



Stores

- Employee ID
- Employee name
- Action type
- Description of activity
- Timestamp

Purpose

- Tracks database changes automatically.
- Helps with auditing and monitoring.

Script 2: Create Deleted Employee Archive Table

```
CREATE TABLE IF NOT EXISTS deleted_employees_archive
```

What it does

Creates a backup/archive table for deleted employee records.

Purpose

- Preserves deleted data before permanent removal.
- Supports recovery and historical reporting.

SECTION 3: INSERT Trigger

Script: Create INSERT Trigger

```
CREATE TRIGGER employee_after_insert  
AFTER INSERT ON employees
```

What it does

Automatically logs employee information whenever a new employee is inserted.

Important Concepts

AFTER INSERT

- Trigger executes after insertion succeeds.



FOR EACH ROW

- Trigger runs once for every inserted row.

NEW

- Refers to newly inserted values.

Example:

```
INSERT INTO employees (...)  
VALUES (...);
```

What It Does

- Inserts a new employee record.

Result

- Trigger automatically runs.
- Employee data is copied into employee_audit_log.

View Audit Log

```
NEW.first_name
```

Means:

- Access the inserted employee's first name.

What Happens Internally

When a new employee is added:

1. Record enters employees
2. Trigger fires automatically
3. Audit log record is inserted into employee_audit_log



SECTION 4: Testing INSERT Trigger

Script: Insert New Employee

```
INSERT INTO employees
```

What it does

Adds a sample employee record.

Purpose

- Tests whether the INSERT trigger works correctly.

Verification Queries

```
SELECT * FROM employee_audit_log
```

What it does

Checks whether the trigger created an audit log entry.

SECTION 5: UPDATE Trigger

Script: Create UPDATE Trigger

```
CREATE TRIGGER employee_after_update  
AFTER UPDATE ON employees
```

What it does

Automatically records employee updates.

Purpose

Tracks modifications made to employee records.



Important Concepts

OLD

Refers to previous values before update.

Example:

```
OLD.department_id
```

NEW

Refers to updated values after modification.

Example:

```
NEW.department_id
```

What Happens Internally

When employee data changes:

1. Record is updated
2. Trigger activates automatically
3. Old and new values are compared
4. Change is logged into audit table

SECTION 6: Testing UPDATE Trigger

Script: Update Employee Department

```
UPDATE employees  
SET department_id = 3
```

What it does

Changes employee department.

Purpose

Tests UPDATE trigger functionality.



Verification Query

```
SELECT * FROM employee_audit_log
```

What it does

Displays update activity logs.

SECTION 7: DELETE Trigger

Script: Create DELETE Trigger

```
CREATE TRIGGER employee_before_delete  
BEFORE DELETE ON employees
```

What it does

Archives employee data before deletion.

Purpose

Prevents permanent data loss.

Important Concepts

BEFORE DELETE

Trigger executes before record removal.

OLD

Contains values about to be deleted.

What Happens Internally

When an employee is deleted:

1. Trigger activates first
2. Employee data is copied to archive table
3. Original record gets deleted



SECTION 8: Testing DELETE Trigger

Script: Delete Employee

```
DELETE FROM employees  
WHERE employee_id = 200;
```

What it does

Deletes employee record.

Purpose

Tests DELETE trigger behavior.

Verification Query

```
SELECT * FROM deleted_employees_archive
```

What it does

Checks whether deleted employee data was archived.

SECTION 9: View Complete Audit History

Script

```
SELECT *  
FROM employee_audit_log  
ORDER BY action_timestamp DESC;
```

What it does

Displays all trigger-generated logs.

Purpose

Provides activity history and tracking.



Key Trigger Concepts

Concept	Meaning
Trigger	Automatic SQL program
INSERT Trigger	Runs after insertion
UPDATE Trigger	Runs after update
DELETE Trigger	Runs before/after deletion
NEW	Newly inserted/updated values
OLD	Previous existing values
FOR EACH ROW	Runs once per affected row

Difference Between Trigger Types

Trigger Type	Executes When
INSERT Trigger	New record added
UPDATE Trigger	Existing record modified
DELETE Trigger	Record removed

Benefits of Triggers

Triggers help with:

- Automation
- Audit logging
- Data validation
- Security monitoring
- Backup creation
- Business rule enforcement
- Historical tracking



Practical Business Use Cases

INSERT Trigger

Used for:

- User registration logging
- Employee onboarding tracking
- Automatic notifications

UPDATE Trigger

Used for:

- Salary changes tracking
- Department transfer logs
- Compliance monitoring

DELETE Trigger

Used for:

- Soft-delete backup systems
- Archive management
- Recovery support

Overall Summary

The scripts demonstrate a complete trigger workflow in MySQL using the TechSphere Solutions database.

The system automatically:

1. Logs new employee insertions
2. Tracks employee updates
3. Archives deleted employee records
4. Maintains full audit history

This creates a more secure, automated, and enterprise-ready database system.